Exercise 1 :

You are given the following four domains:

i. A group of people playing poker.
ii. A person driving a car.
iii. A machine detecting chocolate bars weighing less than 50g.
iv. A doctor performing a medical diagnosis.

Classify each of the domains above along domain the properties that we discussed in the lecture by entering yes/no in the cells of the following table:

	Accessible	Deterministic	Episodic	Static	Discrete	Single agent
Poker						
Car						
Machine						
Doctor						

Solution:

	Accessible	Deterministic	Episodic	Static	Discrete	Single agent
Poker	*no*	*no*	*no*	*yes*	*yes*	*no*
Car	*no*	*no*	*no*	*no*	*no*	*no*
Machine	*yes*	*yes*	*yes*	*yes*	*yes*	*yes*
Doctor	*no*	*no*	*no*	*no*	*no*	*yes*

Exercise 2 :

For each of the following assertions, say whether it is true or false and give a short explanation in 1 or 2 sentences for each of your answers.

(a) An agent that senses only partial information about the state cannot be perfectly rational.

Solution: False. Perfect rationality refers to the ability to make good decisions given the sensor information received.

(b) There exist task environments in which no pure reflex agent can behave rationally.

Solution: True. A pure reflex agent ignores previous percepts, so cannot obtain an optimal state estimate in a partially observable environment. For example, correspondence chess is played by sending moves; if the other player's move is the current percept, a reflex agent could not keep track of the board state and would have to respond to, say, "a4" in the same way regardless of the position in which it was played.

(c) There exists a task environment in which every agent is rational.

Solution: True. For example, in an environment with a single state, such that all actions have the same reward, it doesn't matter which action is taken. More generally, any environment that is reward-invariant under permutation of the actions will satisfy this property.

(d) The input to an agent program is the same as the input to the agent function.

Solution: False. The agent function, notionally speaking, takes as input the entire percept sequence up to that point, whereas the agent program takes the current percept only.

(e) It is possible for a given agent to be perfectly rational in two distinct task environments.

Solution: True. For example, we can arbitrarily modify the parts of the environment that are unreachable by any optimally acting agent as long as they stay unreachable.

(f) Every agent is rational in an unobservable environment.

Solution: False. Some actions are stupid—and the agent may know this if it has a model of the environment—even if one cannot perceive the environment state.

(g) A perfectly rational poker-playing agent never loses.

Solution: False. Unless it draws the perfect hand, the agent can always lose if an opponent has better cards. This can happen for game after game. The correct statement is that the agent's expected winnings are non-negative.

(h) Suppose an agent selects its next action uniformly at random from a set of possible actions. There exists a deterministic task environment in which this agent is rational.

Solution: True. Imagine an environment with only one action.